

I hereby apply for the following recognitions:

Courses from the previous university				Corresponding course at TUM		
Module Nr	Titel	Credit Points	Original Grade	Module Nr	Titel	ECTS
CSE1300	Reasoning and Logic	5	9.5	INHN0004	Discrete Structures	8
CSE1400	Computer Organization	5	10	INHN0003	Introduction to Computer Organization and Tec	8
CSE1305	Algorithms and Data Structures	5	9	INHN0008	Fundamentals of Algorithms and Data Structure	6
CSE1205	Linear Algebra	5	9	INHN0009	Computational Mathematics I: Linear Algebra	8
CSE1200	Calculus	5	10	INHN0014	Computational Mathematics II: Calculus	8
CSE1210	Probability Theory and Statistics	5	9.5	INHN0020	Discrete Probability Theory	6
CSE1500 +	Web and Database Technology + Infor	5+5=1	7.5, 9	INHN0011	Fundamentals of Databases	6
CSE1100	Object Oriented Programming	5	8	INHN0001	Introduction to Informatics	6
CSE1100 +	Object Oriented Programming + Algorit	5+5=1	8, 9	INHN0002	Fundamentals of Programming	6
CSE1405	Computer Networks	5	8	INHN0012	Computer Networking and IT Security	6

Date: 4 September, 2024

Signature:

Higher education system in the Netherlands

Higher education in the Netherlands is organised around a 3-cycle degree system, consisting of bachelor's, master's and PhD degrees. Two types of higher education programmes are offered: research-oriented degree programmes offered primarily by research universities, and professional higher education programmes offered primarily by universities of applied sciences.

Primary and secondary education

Access to higher education

Children are allowed to begin school at the age of 4, but are not legally required to do so until the age of 5. Primary education lasts 8 years (of which 7 are compulsory). During their last year, pupils are advised on the type of secondary education they should pursue.

Secondary education, which begins at the age of 12 and is compulsory until the age of 16, is offered in various forms and at different levels. Vmbo programmes (4 years) combine general and vocational education and prepare pupils to go on to senior secondary vocational education and training (mbo), lasting 1 to 4 years. There are 2 types of general education that grant admission to higher education: havo (5 years) and vwo (6 years). Pupils are enrolled according to their ability. The last 2 years of havo and the last 3 years of vwo are referred to as the '2nd phase' (*2e fase*), or upper secondary education. During these years, pupils focus on 1 of 4 subject clusters (*profielen*), each of which emphasises a certain field of study in addition to satisfying the general education requirements. Each cluster is designed to prepare pupils for study at the tertiary level. A pupil enrolled at a vwo or havo school can choose from the following subject clusters:

1. Science and Technology (*Natuur en Techniek*)
2. Science and Health (*Natuur en Gezondheid*)
3. Economics and Society (*Economie en Maatschappij*)
4. Culture and Society (*Cultuur en Maatschappij*)

Only the 6-year vwo diploma grants access to bachelor's programmes at research universities; the vwo diploma, havo diploma and the highest level of mbo grant access to bachelor's programmes at universities of applied sciences.

Higher education

Higher education in the Netherlands is offered at 2 types of institutions: research universities and universities of applied sciences. Research universities include general universities, universities specialising in engineering and agriculture, and the Open University. Universities of applied sciences include general institutions as well as institutions specialising in a specific field such as agriculture, fine and performing arts or teacher training.

Whereas research universities are primarily responsible for offering research-oriented programmes, universities of applied sciences are primarily responsible for offering programmes of higher professional education, which prepare students for specific professions. These tend to be more practice oriented than programmes offered by research universities.

In this binary, 3-cycle system, bachelor's, master's and PhD degrees are awarded. Short cycle higher education leading to the associate's degree is offered by universities of applied sciences. Degree programmes and periods of study are quantified in terms of the ECTS credit system.

The focus of degree programmes determines both the number of credits required to complete the programme and the degree which is awarded. A research-oriented bachelor's programme requires the completion of 180 credits (3 years) and graduates obtain the

degree Bachelor of Arts, Bachelor of Science, or Bachelor of Law (BA/BSc/LLB), depending on the discipline. In most cases, a bachelor's degree awarded in the applied arts and sciences requires 240 credits (4 years), to complete. The majority of students obtain a BA/BSc/or LLB degree, but those graduating from programmes in business administration, social work, education and music may obtain a BBA/BSW/BEd or BM, respectively. An increasing number of universities of applied sciences are offering 3-year bachelor's degree programmes for students with a vwo diploma. These programmes are both intensive and challenging, enabling students to complete a bachelor's programme in the applied arts and sciences after completion of 180 credits. An associate's degree (Ad) in the applied arts and sciences requires 120 credits (2 years). The Ad can be taken either as a terminal degree, after which graduates can seek employment, or students who complete the 2-year programme can continue studying for a bachelor's degree in the applied arts and sciences.

A research-oriented master's programme requires the completion of 60, 90 or 120 credits (1, 1.5 or 2 years). In engineering, agriculture, and mathematics and the natural sciences, 120 credits are always required. Graduates obtain a Master of Arts, Master of Science, or Master of Laws (MA/MSc/LLM). A master's degree awarded in the applied arts and sciences requires the completion of 60 to 120 credits. The majority of students obtain an MA/MSc/or LLM degree, but those graduating from programmes in business administration, social work, education and music may obtain an MBA/MSW/MEd or MM, respectively.

The 3rd cycle of higher education, leading to a PhD or to a Professional Doctorate in Engineering (PDEng), is offered only by research universities. The major requirement for the PhD, which is offered by all research universities, is completion of a dissertation based on

original research that is publicly defended. In addition to PhD programmes, the 3 engineering universities offer technological design programmes consisting of advanced study and a personal design assignment in a number of engineering fields. The technical design programme requires 2 years of study to complete and graduates obtain the degree

Professional Doctorate in Engineering (PDEng). The training of medical specialists is the responsibility of the professional group in an organisational setting at a university hospital.

Requirements for access to higher education

For access to research-oriented bachelor's programmes, students are required to have a vwo diploma or to have completed the 1st year (60 credits) of a bachelor's programme at a university of applied sciences. For the latter category of students, additional selection criteria may apply. The minimum access requirement to universities of applied sciences is either a vwo diploma, a havo diploma or a diploma of secondary vocational education (mbo), provided certain conditions are met. The vwo diploma not only grants access to universities of applied sciences, but based on this diploma, students may receive exemption from 1 year of study as well.

For access to both types of higher education, pupils with a vwo or havo diploma are required to have completed at least 1 of the subject clusters that fulfil the requirements for the higher education programme in question. A *quotum*, or *numerus fixus*, applies for access to certain programmes, primarily in the medical sciences. For *numerus fixus* programmes, institutions are permitted to select the students they admit based on academic performance, personal motivation, etc. Potential students older than 21 years who do not possess one of the qualifications mentioned above can qualify for access to higher education on the basis of an entrance examination and assessment (recognition of prior learning). For access to certain programmes, particularly those in the fine arts, students have to demonstrate the required artistic abilities. The only access requirement for the Open University is that applicants be at least 18 years of age.

For access to all master's programmes, a bachelor's degree in 1 or more specific disciplines is required, in some cases in combination with other requirements. Graduates with a bachelor's degree in the applied arts and sciences usually have to fulfil additional requirements for admission to a research-oriented master's programme.

Credit system and grading

A student's workload is measured in ECTS credits. According to Dutch law, 1 credit represents 28 hours of work and 60 credits represents 1 year of full-time study. The grading system used in the Netherlands is on a scale from 1 (very poor) to 10 (outstanding). The lowest passing grade is 6; 9s are seldom given and 10s are extremely rare. Grades 1-3 are hardly ever used. The academic year is 42 weeks long.

Quality assurance and accreditation

A guaranteed standard of higher education, and alignment with the Qualifications Framework for the European Higher Education Area, is maintained through a system of legal regulation and quality assurance, in the form of accreditation. The Ministry of Education, Culture and Science is responsible for legislation pertaining to education. The agriculture and public health ministries play an important role in monitoring the content of study programmes in their respective fields.

Quality assurance is carried out through a system of accreditation, administered by the [Accreditation Organisation of the Netherlands and Flanders \(NVAO\)](#). According to the Dutch Higher Education Act, all degree programmes offered by research universities and universities of applied sciences must be evaluated according to established criteria. Programmes that meet the criteria are accredited: i.e. recognised for a period of 6 years. Only accredited programmes are eligible for government funding; students receive financial aid and graduate with a recognised degree only when enrolled in, and after having completed, an accredited degree programme. All accredited programmes are listed in the Central Register of Higher Education Study Programmes (CROHO).

As part of the accreditation system, higher education institutions can request the NVAO to conduct an 'institutional quality assessment' to determine the extent to which the institution is capable of guaranteeing the quality of the programmes it offers. Programmes offered by institutions that receive a positive evaluation still have to be accredited, but the accreditation procedure takes less time and is not as extensive.

Besides the accreditation of degree programmes, the Netherlands has a system by which the Ministry of Education, Culture and Science recognises higher education institutions by conferring on them the status of either 'funded' or 'approved'. "Funded" indicates the institution is fully financed by the government. "Approved" indicates that the institution does not receive funds from the government and has to rely on its own sources of funding.

Whether a degree programme is offered by a 'funded' or an 'approved' institution, it must be accredited and registered in CROHO to be considered recognised.

Please note: if a bachelor's or master's degree programme is not registered in CROHO, the quality is not assured by the Dutch quality assurance system. The quality may however be assured by another system.

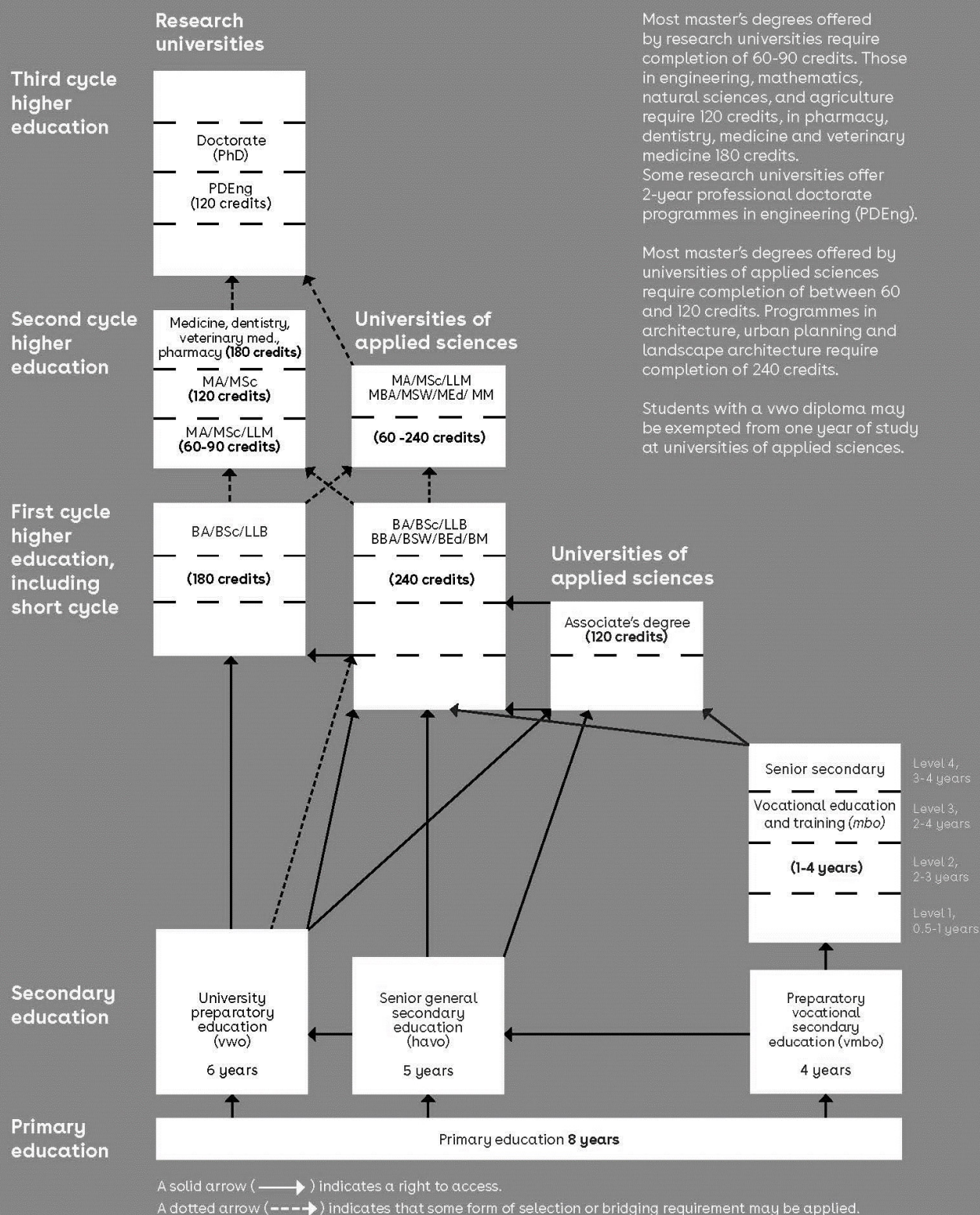
National Qualifications Frameworks

An important tool to facilitate the recognition of foreign qualifications is using overarching qualifications frameworks as a translation tool through which qualifications awarded in one country can be compared to qualifications awarded abroad. A comprehensive overarching framework used in the European Economic Area is the European Qualifications Framework for Lifelong Learning (EQF-LLL). The EQF-LLL describes the learning outcomes associated with qualifications at eight different levels and is used as a common reference framework to assist in comparing national qualifications systems and their levels. The qualifications framework in the Netherlands is referred to as the Dutch Qualifications Framework (NLQF). The NLQF was officially referenced to the EQF in 2012. The NLQF has a total of nine levels: an "entry level"

which is below level 1 of the EQF-LLL and therefore not referenced to the EQF-LLL, and 8 levels which are referenced to the 8 levels of the EQF. You can find more information on the Dutch Qualifications Framework on the website of the [National Coordination Point NLQF](#), which is the organisation responsible for the development and implementation of the NLQF.

The Dutch education system

The higher education system in the Netherlands is based on a three-cycle degree system, consisting of a bachelor, master and PhD. Two types of programmes are offered: research-oriented degree programmes offered by research universities, and professional higher education programmes offered by universities of applied sciences.



Grading scheme TU Delft BSc students

Dutch grading scale

The Dutch grading scale, implemented from primary school through to university level, goes from 1 (very poor) to 10 (excellent). At TU Delft final marks are rounded off to whole and half figures. A 6,0 is a pass. Although the scale runs from 1,0 to 10,0, the marks 9,0 and 9,5 are not frequently awarded and 10s are extremely rare. The meaning of the marks is as follows:

Mark	Definition
9,5 - 10,0	Excellent
8,5 - 9,0	Very good
7,5 - 8,0	Good
6,5 - 7,0	More than satisfactory
6,0	Satisfactory
4,5 - 5,5	Nearly satisfactory
3,5 - 4,0	Unsatisfactory
1,0 - 3,0	Very poor

Sometimes a 'V' is awarded instead of a mark to indicate that the requirements for a particular subject have been fulfilled.

'VR' instead of a mark indicates that a student has been exempted from an examination or practical exercise. The board of examiners may decide to exempt students from an examination or practical exercise on the grounds of a previous examination, degree audit or practical exercise completed within the higher education system that corresponds with the subject for which exemption is sought, as regards content and study load, or on the grounds of knowledge and/or skills acquired outside the higher education system.

Average results of TU Delft BSc students

Grading systems and 'grading culture' vary from country to country. In the Netherlands a 1 – 10 grading scale is commonly used. The Dutch grading culture is rather conservative: a 10, or even a 9, is not often awarded. At TU Delft even higher standards are the norm.

BSc average grading

For a better appreciation of TU Delft BSc student results, the overview below shows the distribution of grades obtained by students graduating from September 2019 to August 2022.

Grade	% obtained	% better scores
10	3.6	0.0
9	19.4	3.6
8	39.3	23.0
7	28.2	62.3
6	9.3	90.5
5	0.2	99.8

The ECTS grading system for credit transfer

In cases where there is a need for international credit transfer (mainly in conjunction with student exchanges) *ECTS grades* can be used. It is good practice to award an ECTS grade, particularly if credit transfer is later required.

The ECTS grading system is based on the relative distribution of the results of the students who have passed an examination. Grades are assigned as follows among students with a pass:

Grade	Relative distribution
A	Best 10 %
B	Next 25 %
C	Next 30 %
D	Next 25 %
E	Next 10 %
FX/F	Fail

A distinction is made between the grades *FX* and *F* which are given to unsuccessful students. *FX* means that in order to pass more work is required and *F* means that considerable further work is required.